MILITARY SPECIFICATION SHEET

ELECTRON TUBE, KLYSTRON

1/ TYPE 8848

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Reflex, integral cavity, mechanically tuned, 8,450 to 10,050 MHz, waveguide output.

ABSOLUTE RATINGS:

				Temperature			
Parameter:	Ef	Ers	Er	Body	Base	Cap	Alt
Unit:	V	Vdc	Vdc	°C	°C	°C	ft
Maximum:	7.0	400	-1,000	160	120	120	10,000
Minimum:	5.7	---	-20	---	---	---	---

PHYSICAL CHARACTERISTICS:

Dimensions: See figure 1
Cathode: Coated, unipotential
Mounting position: Any
Weight: 6 ounces (170.1 grams)

RF connection: Waveguide, mates with UG-39U (MIL-F-3922/53)
Base: Pee wee, 3 pin (A3-1)
Cap: Miniature (C1-4)

TEST CONDITIONS: See note 1

Parameter:	Ef	Ers	Er	F	T (body)
Unit:	V	Vdc	Vdc	MHz	°C
	6.3	350	-55 to -225	F2	100 (max)

	FREQUENCY	
F	MHz	Tolerance (MHz)
F1	8,450	+00, -10
F2	9,300	+10
F3	10,050	+10, -00

GENERAL:

Qualification - Not required

Formerly V-262/MXK-48

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 1						
4214	Cathode emission	2	Ef = 5.7 V	ΔIk/Ik	---	-15	%
1256	Electrode current (resonator)	-	Er/max Po	Irs	---	42	mAdc
4213	Reflector voltage	-	Er/max Po; F = F1 and F3	Er	-150	-300	Vdc
4229	Total reflector current	2, 3		Ir	---	5	μAdc
4223	Mechanical tuning range	-	Er/max Po				
			Low frequency	F	---	8,450	MHz
			High frequency	F	10,050	---	MHz
4289	Heater current (nonoperating)	-	Ef = 6.3 V	If	1.08	1.32	A
4250	Power output	-	Er/max Po; F = F1 thru F3	Po	65	---	mW
	Quality conformance inspection, part 3						
---	Life test	-	Group D	t	500	---	hrs
---	Life-test end point:						
4250	Power output	-	Er/max Po; F = F1 thru F3	Po	52	---	mW
4223	Tuning rate	4	F = F1 thru F3	---	---	740	Degrees
4223	Operating torque	4	F = F1 thru F3	---	---	4	in.-lbs
4027	Temperature coefficient	4	Er/max Po; TA = 20° to 60°C	---	-250	+250	kHz/°C
1031	Nonoperation vibration	4	t = 120 minutes, 40 minutes in each of three planes X, Y, and Z	---	---	---	---

NOTES:

1. All oscillation tests except vibration tests shall be made with the tube rigidly connected to a UG-39/U (MIL-F-3922/53) flange on appropriate RG-52/U waveguide equipment and the load VSWR for the tube shall be less than 1.1:1. Forced-air cooling is required for power inputs above 10 watts. For maximum tube life, the tube temperature should be less than 100°C.
2. The tube shall not be oscillating during test.
3. After 2 minutes with all voltages applied, total reflector current shall not exceed the specified limits.
4. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.
5. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 80
DSA - ES

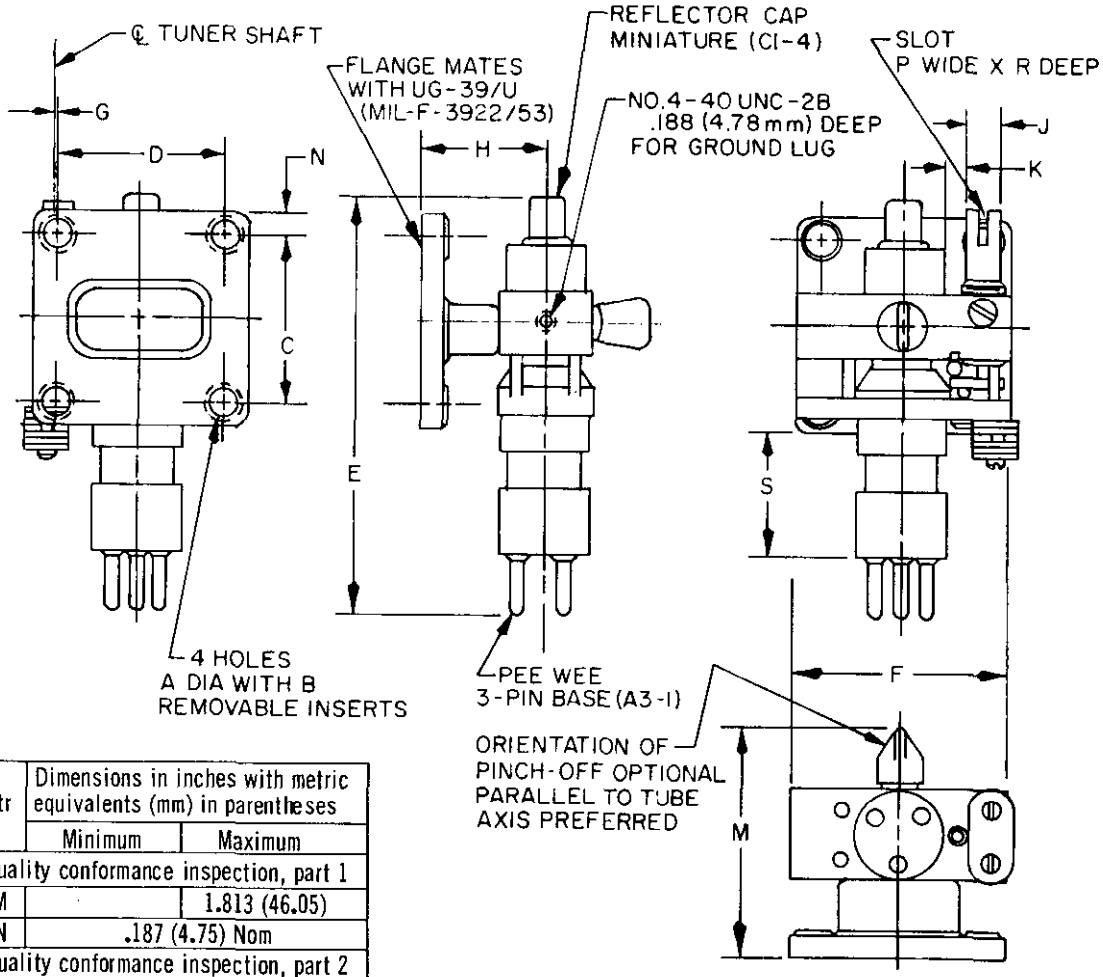
User activities:
Army - WC, ME
Navy - AS, OS, MC, CG
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-3190)

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 1		
M		1.813 (46.05)
N	.187 (4.75) Nom	
Quality conformance inspection, part 2		
A	.219 (5.56) Nom	
B	.185 (4.70) Nom	
C	1.276 (32.41)	1.284 (32.61)
D	1.216 (30.89)	1.224 (31.09)
E		3.125 (79.38)
G	.010 (.25)	.020 (.51)
H	.850 (21.59)	.936 (23.77)
J	.248 (6.30)	.250 (6.35)
P	.050 (1.27) Nom	
R	.220 (5.59) Nom	
Quality conformance inspection, part 3 (see note 4)		
F		1.750 (44.45)
K	.160 (4.06)	
S		1.000 (25.40)

PIN CONNECTIONS		
1	h	} int con
2	k	
3	h	

FIGURE 1. Outline drawing of electron tube type 8848.

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